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Gender difference in hepatic AMPK pathway activated lipid metabolism induced by aged polystyrene microplastics exposure

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ABSTRACT

Microplastics (MPs) pollution becomes an increasing concern and researchers keep exploring the health effects caused by MPs exposure. The ageing process in the environment significantly alters the physicochemical characteristics of MPs and subsequently affects their toxicities. The health effects of aged MPs exposure and the mechanism underlying are worthy of exploration. Polystyrene microplastics (PS-MPs) (with size less than 50 μm) were obtained by grinding and screening polystyrene materials. PS-MPs continued to be aged by ozone treatment (0.4 mg/min, 9 h). Both male and female C57BL/6 mice were orally exposed to 0 or 2 mg/kg/d aged PS-MPs for 28 days. Results showed that PS-MPs were found in liver, ovary and spleen of females and liver, testis and spleen of males in the aged PS-MPs group. Exposure to aged PS-MPs significantly decreased abdominal fat/body coefficient, the adipocyte size and the serum LDL-C level in females. Compared to the control, serum estradiol (E2) level, the mRNA expression levels of genes regulating E2 production (*17β-hsd*, *3β-hsd* and *Star*) in ovary and the protein expression levels of E2 receptors (ERα, ERβ), AMPKα and p-AMPKα1 in liver increased significantly, and the mRNA expression levels of AMP-activated protein kinase (AMPK) downstream genes (*Srebp-1c*, *Fas* and *Scd1*) in liver decreased significantly in the female aged PS-MPs group. Liver metabolomic profiling showed that differential metabolites between female aged PS-MPs group and female control group were enriched in biotin metabolism and the level of biotin increased significantly in the female aged PS-MPs group. However, no significant changes were detected in males. These results indicated that aged PS-MPs exposure increased ovarian E2 production and activated the AMPK pathway in the liver which might inhibit liver lipid synthesis only in females. Our findings provide new insights into the potential sex-specific health effects of environmental MPs pollution.

1. Introduction

The production and consumption of plastics are constantly increasing due to their great demand in both industries and daily lives (Li et al., 2018). Accordingly, around 10 million metric tons of plastics enter the ocean yearly (Landrigan et al., 2020) where plastics can be decomposed into small pieces under external conditions with time in the environment including mechanical action, biodegradation, ultraviolet radiation and oxidation (Qi et al., 2018; Liu et al., 2019a). The plastic pieces smaller than 5 mm were defined as MPs (Thompson et al., 2004).

MPs are commonly found in various environmental media including atmosphere, water and soil (Xiong et al., 2018; Xu et al., 2020; Gaston et al., 2020) and are easy to be ingested by organisms (Xiong et al., 2018; Xu et al., 2020; Steer et al., 2017). Some studies found that MPs could be transferred and accumulated in the liver after entering the body (Lu et al., 2018; Deng et al., 2017; Zheng et al., 2021). Furthermore, MPs were also found to affect hepatic lipid metabolism although the underlying mechanism is still being explored (Lu et al., 2018).

The AMP-activated protein kinase (AMPK) signaling pathway is a widely recognized lipid-producing signal (Carling, 2004). Studies have shown that AMPK could prevent fat-synthesis-related targets, such as sterol regulatory element-binding protein 1c (*Srebp-1c*), acetyl-CoA carboxylase (*Acc*), fatty acid synthase (*Fas*) and stearoyl-CoA desaturase1 (*Scd1*), and promote the fatty-acid-oxidation-related proteins, such as carnitine palmitoyl transferase1 (*Cpt1*) (Fan et al., 2018; Guo et al., 2020). AMPK pathway could be activated by biotin and hormones such as the predominant circulating estrogen estradiol (E2) in the liver (Aguilera-Méndez and Fernández-Mejía, 2012; Pedram et al., 2013). The

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key elements in the AMPK pathway and their regulators could be the potential targets for the internal MPs.

Another important issue needing to be emphasized in the MPs toxicological studies is the materials exposed. In recent studies focusing on MPs health effects, pristine MPs were mostly used as exposure materials which could not represent the real MPs in the environment (Guimarães et al., 2021; Liu et al., 2020). MPs experience ageing process in the environment which alters their physical properties such as becoming more hydrophilic, having a larger surface area, more oxygen-containing functional groups and stronger adsorption capacities (Salehi et al., 2018). Scientists proved that UV radiation could increase toxicity of polystyrene microbeads in fish and *Caenorhabditis elegans* (Chen et al., 2021a; Wang et al., 2020a; Chen et al., 2022). It would be more interesting to investigate the health effects induced by aged MPs from real plastic materials since it would be closer to actual MPs exposure.

Polystyrene (PS) is one of the most commonly used plastics in the world (Barnes et al., 2009). Therefore, in our study, PS-MPs were made from PS products, aged with ozone and exposed to female and male mice. PS-MPs were detected in mice organs. The production of E2, hepatic expressions of E2 receptors and key elements in the AMPK pathway were measured, and liver metabolomics profiling was analyzed to investigate the effects of aged PS-MPs exposure on hepatic lipid metabolism in mice.

2. Materials and methods

2.1. Materials

PS-MPs were prepared by crushing and grinding PS plates (Luancheng District Guangsai Insulation Material Factory, Shijiazhuang, China) with a plastic crusher (FW100, Taisite Tianjin, China). PS-MPs less than 50 μm in size were obtained after screening with stainless steel screens. Then the PS-MPs were washed with 70% ethanol three times and with deionized water twice. This kind of PS-MPs was called grinded PS-MPs in this study. After being dried at 50 °C, the PS-MPs were aged in a flask ventilated with ozone at a rate of 0.4 mg/min for 9 h (Ozone Generator, OGP-S910GC, Opfa Guangzhou, China) (Miranda et al., 2021). Aged PS-MPs were collected and stored at room temperature. In the animal study, aged PS-MPs were dispersed evenly in the corn oil (Chen et al., 2022).

2.2. Characterization of aged PS-MPs

Both grinded and aged PS-MPs infrared spectra were obtained with a Fourier Transform Attenuated Total Reflection Infrared Spectroscope (FTIR-ATR, Nicolet iS50, Thermo Fisher, USA). The morphologic pictures of grinded and aged PS-MPs were taken with a scanning electron microscope (SEM, JSM-6510, JEOL, Japan). Particle sizes were evaluated in more than 300 PS-MPs with Nano Measure 1.2.0 (Surface Chemistry and Catalysis Laboratory, China).

2.3. Animals

Fourteen female and fourteen male C57BL/6 mice (eight-week-old) (Jiangsu Laboratory Animal Center) were housed in the specific pathogen-free (SPF) laboratory (Nanjing Medical University, SYXK(SU) 2020–0006) on a 12 h light/12 h dark cycle with temperature 20 ± 2 °C and relative humidity 50~60% and with food and water available ad libitum. Animal care and all experiments were approved by the local Animal Care and Use Committee of Nanjing Medical University (IACUC-2006027).

After one week of adaptation, mice were divided into female control group (corn oil), female PS-MPs group, male control group (corn oil) and male PS-MPs group (seven in each group). PS-MPs and control groups were gavaged with 2 mg/kg/d aged PS-MPs (suspended in corn oil) and corn oil respectively for 28 days continuously. The concentration of 2 mg/kg/d was selected according to previous animal studies in literature (Xie et al., 2020; Li et al., 2020; Yang et al., 2019)and in our lab. 2 mg/kg/d PS-MPs exposure could for sure disrupt hepatic lipid metabolism so that the following mechanism could be studied. On the 29th day, male mice were fasted for 8 h and then sacrificed. For female mice, vaginal lavages were obtained using a smear with 0.9% saline (0.9% NaCl solution) and viewed under a microscope daily (×100) since the 29th day. When vaginal cytological result was nucleated which meant estrous status in the coming a few hours, female mice were fasted and then sacrificed. Blood was collected from venae angularis and the serum was obtained by centrifuging blood at 3000 r/m for 15 min at 4 °C. One part of the liver and abdominal fat were fixed with 4% (g/vol) paraformaldehyde. The other part of the liver and abdominal fat and kidney, spleen, ovary/testis tissues, blood serum were stored at -80 °C immediately. The organ coefficients of the mouse were calculated by dividing the organ weight by the weight of the mouse.

2.4. Detection of aged PS-MPs

In this experiment, all reagents were filtered with 0.7 μm membranes (Whatman, UK) before using. Equipment and tools were wiped with ethanol and there were no PS-made tools and containers used in the experiment. Three frozen liver samples were randomly selected from each group and 15 mg liver piece was cut down from each liver and pooled (45 mg totally). Samples were digested with KOH and NaClO solutions at 50 °C for 48 h. The digested sample solutions were filtered with 25 mm quartz membranes (Whatman, UK). The membranes were carefully stored in the membrane box (Haiklas Experimental Equipment Co., Ltd, China). Kidney, spleen and ovary/testis samples were digested the same way. The whole process of sample digestion in this study was carried out in the biosafety cabinet (CBC1100-IIA2, Beijing Donglian Har Instrument Manufacture Co., Ltd, China) to avoid contamination from the environment. Particles on the membranes were detected with the LabRAM HR Evolution Raman Spectrometer (Horiba scientific, France) and each particle spectrum was compared with PS spectrum reference using KnowItAll software (BioRad Laboratories, Inc., USA). PS particles were confirmed when the Hit Quality Index (HQI) was above 80 (Ragusa et al., 2021).

2.5. Blood biochemical analysis

The alanine aminotransferase (ALT), alkaline phosphatase (ALP), aspartate aminotransferase (AST), high density lipoprotein cholesterol (HDL-C), low density lipoprotein cholesterol (LDL-C), triglyceride (TG) and total cholesterol (TCHO) levels in the mice serum were measured at one time with an automatic biochemical analyzer (HITACHI, Tokyo, Japan). Tests were carried out according to the manufacturer's guidelines. The minimum detection limits of ALT, ALP and AST were 2.4, 2.8 and 4 U/L, respectively and the minimum detection limits of HDL-C, LDL-C, TG and TCHO all were 0.1 mmol/L.

2.6. Hematoxylin-eosin (H&E) staining

Liver and abdominal fat tissue fixed with 4% (g/vol) paraformaldehyde were embedded in paraffin and sectioned using HM 340E Rotary Microtome (Thermo Scientific, Wilmington, USA). Sample slices were H&E stained and scanned with the Pathological Scanner (3DHISTECH, Hungary). For each mouse, four fields were randomly selected in each abdominal fat slice and three slices were chosen. The sizes of adipocytes were measured with Image J 1.51J8 (National Institutes of Health, USA) and the average adipocyte size was calculated in each group.

2.7. E2 measurement

Serum E2 levels were measured with the ELISA Kit (Beyotime, China) according to the manufacturer's guidelines. The minimum detection limit was 12 pg/mL. E2 levels of all samples were measured in one assay to avoid inter-batch variation (three replicates).

2.8. RNA extraction and real-time PCR (RT-PCR)

Total RNA was extracted from the livers using Trizol reagent (Takara Biochemicals, China) and reversed to cDNA using a reverse transcriptase kit (Takara Biochemicals, China). RT-PCR was carried out on Light-Cycler 480 (LC480) (Roche, Basel, Switzerland) using the SYBR Green system (Takara Biochemicals, China) with *Gapdh* as the housekeeping gene. The mRNA levels of *Star*, *17β-hsd*, *3β-hsd*, in ovary/testis and *Srebp-1c*, *Acc*, *Fas*, *Cpt1* and *Scd1* in liver were measured (four replicates for each). All the primer sequences of RT-PCR were shown in Supplementary Table S1.

2.9. Western blotting

Total proteins were extracted from the livers and the protein concentrations were detected using the BCA protein concentration determination kit (Beyotime, China). The reaction systems were prepared according to the protein concentration and boiled under 100 °C for 5 min 10% SDS-polyacrylamide gel electrophoresis (SDS-PAGE, EpiZyme, China) was used and proteins were transferred to polyvinylidene fluoride (PVDF) membranes. The membranes were incubated with rabbit anti-GAPDH (Cell Signaling Technology, USA, 1:1000), mouse anti-ERα (Santa Cruz Biotechnology, USA, 1:500), mouse anti-ERβ (Santa Cruz Biotechnology, USA, 1:500), rabbit anti-GPR30 (Immunoway, USA, 1:1000), rabbit anti-AMPKα (Cell Signaling Technology, USA, 1:1000) and rabbit anti-p-AMPKα1 (Cell Signaling Technology, USA, 1:1000) overnight at 4 °C. HRP-labeled goat anti-rabbit/mouse IgG (H+L) (Beyotime, China, 1:1000) was used as a secondary antibody. Protein expressions were analyzed with Image J software according to their grayscale.

2.10. Liver metabolomic profiling

Liver samples (50 mg) were pretreated with 460 μL methanol, 150 μL ultrapure water, 10 μL Internal standard A (Isotop internal standards containing L-phenylalanine acid, Creatinine, Glutarate, Valine N-acetaminophenol, Niacin, Horse uric acid and Thymine), 10 μL Internal standard B (Isotop internal standards containing Decanoic acid), 10 μL Internal standard C (Isotop internal standards containing Tetracosanoic acid). Pretreated samples were centrifuged at 20000 g for 15 min at 4 °C. The supernate was obtained and dried in a centrifugal (Labconco, USA) and resuspended with 20 μL ultra-pure water. The metabolomic profiling was performed with a UPLC Ultimate 3000 system (Dionex, Germering, Germany) combined with a Q-Exactive mass spectrometer (QEMS) (Thermo Fisher Scientific, USA) in both positive and negative modes simultaneously. The UPLC analysis was done as described previously (Chen et al., 2021b). Metabolite pathway analysis was carried out using MetaboAnalyst 5.0 software (Genome Canada, Canada).

2.11. Statistical analysis

All data were represented as mean ± standard deviation (SD) and analyzed using SPSS 25.0 (SPSS, Chicago). Data were tested for normality first. When data were normal, unpaired two-tailed Student's t-test was used for further analysis. Mann-Whitney U test was applied when data were not normal. The value P less than 0.05 were considered statistically significant. GraphPad Prism 7.04 (GraphPad Software Inc., San Diego, CA, USA) was used to make statistical graphs.

3. Results

3.1. Characteristics of aged PS-MPs

The 500–4500 cm^{-1} range of the FTIR-ATR spectrum of grinded and aged PS-MPs was shown in Fig. 1A, which was consistent with the FTIR-ATR spectrum of PS (Cappello et al., 2021; Jeyasanta et al., 2020). The peak assignments and the modes of vibration were shown in Supplementary Table S2. Compared to grinded PS-MPs, stronger and additional peaks characteristic of C═O was observed at wavenumbers of 1717.78 cm^{-1} in aged PS-MPs, indicating more oxidation of PS-MPs after ozone treatment (Zvekic et al., 2022). Although the morphological difference before and after ozone treatment was not as big as reported in other studies using PS microbeads (Lai et al., 2021; Mu et al., 2022), SEM images showed that the aged PS-MPs had rougher surfaces, more irregular shapes and smaller sizes than the grinded PS-MPs (Fig. 1B; Fig. 1D). The size distribution of grinded PS-MPs and aged PS-MPs were shown in Fig. 1C and Fig. 1E. Ozone treatment decreased the average size of PS-MPs from 55 ± 19 μm (grinded) to 27 ± 7 μm (aged). The median value of the PS-MPs size reduced from 53.09 μm (grinded) to 25.94 μm (aged) after ozone treatment.

3.2. Organ coefficients and weight of mice

Compared to the control group, the female aged PS-MPs group had a significantly reduced abdominal fat coefficient ($P < 0.05$) and no significant differences were found between male aged PS-MPs group and male control group (Fig. S1). What's more, the body weight didn't differ between aged PS-MPs and the control groups during the PS-MPs exposure (Fig. S1).

3.3. Detection of aged PS-MPs

PS-MPs were detected in aged PS-MPs group mice. The FTIR-ATR spectra of detected aged MPs in mice confirmed the material of PS (Han et al., 2021; Im et al., 2021). Specifically, in females, one PS-MPs (43.49 μm in size) was detected in pooled liver pieces, one PS-MPs (41.40 μm) was found in pooled ovary pieces and one (58.18 μm) in the pooled spleen pieces. In male aged PS-MPs group, one PS-MPs (48.46 μm) was detected in pooled liver pieces, one (39.70 μm) was found in pooled testis pieces and five (81.6, 49.65, 25.05, 33.94 and 48.05 μm) in pooled spleen pieces. No PS-MPs were detected in female and male pooled kidney pieces and control group tissues. All detected PS-MPs were off-white and irregularly shaped (Fig. 2).

3.4. Blood biochemical analysis

Blood biochemical results showed that compared to the control, female aged PS-MPs treated group serum LDL-C level decreased significantly ($P < 0.05$). In the male mice, no significant differences were found (Fig. 3).

3.5. H&E staining of liver and abdominal fat

No obvious inflammation and lipid droplets were observed in the stained liver in the aged PS-MPs mice (supplementary Fig. S2.). Statistical analysis showed that the adipocyte size in the female aged PS-MPs group decreased significantly compared with the control group ($P < 0.01$) (Fig. 4). However, no significant differences were found in males.

3.6. The serum E2 level, the mRNA expression levels of Star, 17β-hsd and 3β-hsd in ovary/testis and the protein expression levels of ERα, ERβ and GPR30 in liver

The female aged PS-MPs group had significantly higher serum E2

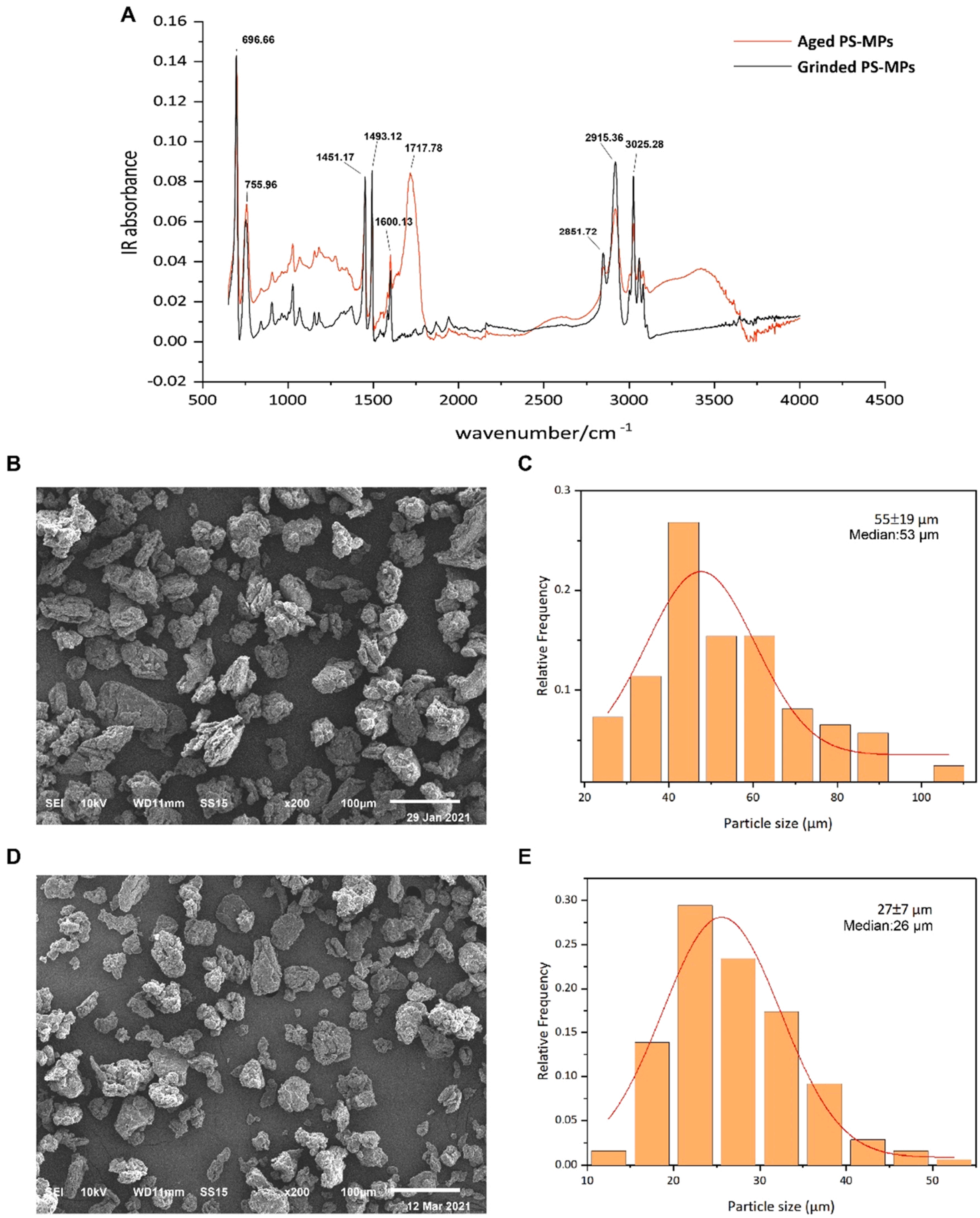

Fig. 1. Characterization of grinded and aged PS-MPs(A) The FTIR-ATR spectra of both grinded and aged PS-MPs in the range of 500–4500 cm^{-1}. (B) The representative SEM image of grinded PS-MPs. (C) The size distribution of grinded PS-MPs. (D) The representative SEM image of aged PS-MPs. (E) The size distribution of aged PS-MPs.

concentration compared to the control ($P < 0.05$) while serum E2 concentration in the male aged PS-MPs group did not change significantly (Fig. 5A). After being exposed to aged PS-MPs, the mRNA expression levels of genes regulating E2 production (*Star*, *17β-hsd* and *3β-hsd*) in ovary increased significantly (all $P < 0.05$) (Fig. 5B). No difference was found in mRNA expression levels of *Star*, *17β-hsd* and *3β-hsd* in males after PS-MPs exposure (Fig. S3). Compared to the control, the protein levels of ERα, ERβ in the female aged PS-MPs group increased significantly ($P < 0.05$, $P < 0.01$), but no significant change was found in the protein expression of GPR30 in liver (Fig. 5C). However, in the male aged PS-MPs group no significant changes were found in the protein levels of ERα, ERβ and GPR30 in liver compared to the control (Fig. 5D).

3.7. The protein expression levels of AMPKα and p-AMPKα1 and the mRNA expression levels of Srebp-1c, Acc, Fas, Cpt1 and Scd1 in liver

Compared to control, the protein expression levels of AMPKα and p-AMPKα1 in liver were up-regulated significantly in the female aged PS-MPs group ($P < 0.001$, $P < 0.01$, respectively) (Fig. 6A), but not in the males (Fig. 6B). Compared to the corn oil group, the mRNA expression levels of AMPK pathway downstream genes (*Srebp-1c*, *Fas* and *Scd1*) in liver decreased significantly in the female aged PS-MPs group ($P < 0.05$,

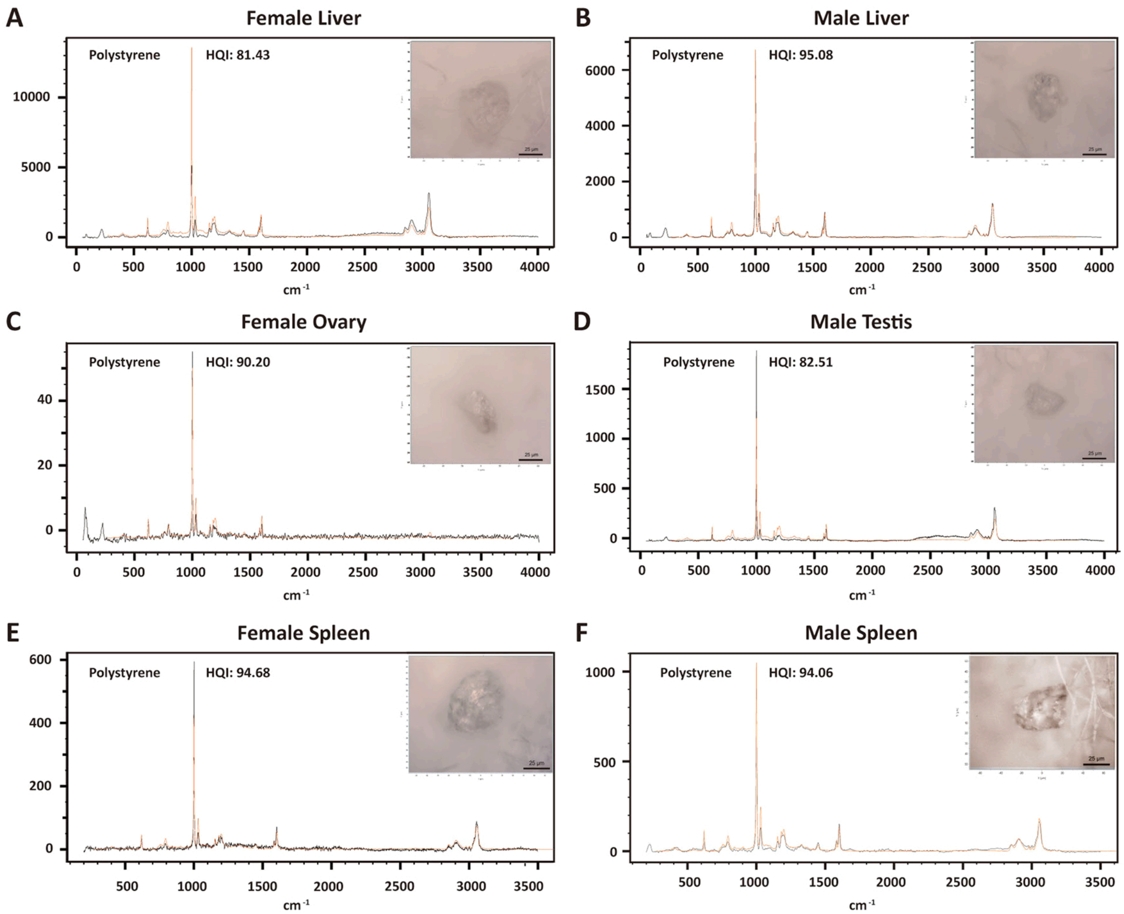

Fig. 2. Raman spectra and the pictures of representative aged PS-MPs detected in female liver (A), male liver (B), female ovary (C), male testis (D), female spleen (E) and male spleen (F).

$P < 0.01$ and $P < 0.05$, respectively) (Fig. 6C), but no significant change was found in *Cpt1*. No significant differences were found between male aged PS-MPs group and male control group (Fig. 6D).

3.8. *UPLC-MS metabolomic profiling of liver*

Liver metabolomic profiling results showed that five differential metabolites (2-Piperidinemethanol, 3'-AMP, Biotin, N-Acetylleucine and Purine) were identified as different between female PS group and female control group and eleven differential metabolites (2-Hydroxycaproic acid, 2-Piperidinemethanol, 3-Methylhistidine, 3-Pyridylacetic acid, 6-Hydroxynicotinic acid, Hyodeoxycholic acid, L-Leucine, Norvaline, N-Acetylvaline, Ursodeoxycholic acid and Xanthosine) were different between male PS group and male control group (Table 1). Through pathway analysis based on the SMPDB database, differential metabolites in the female mice were enriched in biotin metabolism and the level of biotin was upregulated significantly in the female aged PS-MPs ($P < 0.05$). No metabolism pathway was found to enrich the differential metabolites in male mice.

4. Discussion

This study firstly reported a sex-specific effects in liver lipid metabolism in mammals caused by PS-MPs oral exposure. The results indicated that the direct detection of PS-MPs in the ovary was connected with increased production and release of E2 which next upregulated hepatic estrogen receptors expression and therefore activated AMPK pathway to inhibit lipid synthesis. Increased biotin level by PS-MPs in females was also thought to promote the AMPK pathway activity. Pieces from endocrine system and digestive system were linked to join lipid metabolic network to decipher how oral exposure of PS-MPs could decrease the lipid synthesis.

Currently, MPs are regarded as a novel kind of pollutants and there are increasing studies on MPs accumulation in aquatic organisms, but toxicities of MPs in mammals is still under exploration (Li et al., 2018; Sun et al., 2021; Hou et al., 2021; Botterell et al., 2019; Compa et al., 2018). Due to their persistent and degradation-resistant properties, MPs entering the food chain (Miranda and de Carvalho-Souza, 2016) could be enriched in the higher nutritional level, leading to bioaccumulation and biomagnification in mammals especially humans (Lv et al., 2019; Tang et al., 2021). Up to date, MPs were found in gastrointestinal tract, liver, kidney, heart, spleen, lung, brain, testis, ovary and the placenta in mammals and the sizes of detected MPs in the organisms were mainly less than 50 μm (Xu et al., 2020; Deng et al., 2017; Jin et al., 2021; Liu et al., 2022; Tian et al., 2019; Singh and Krishna, 2010; Yao et al., 2020). Philipp and his colleagues detected MPs in human faeces (Schwabl et al., 2019), indicating that it is urgent to investigate the health effects of MPs exposure. In most published studies exploring toxicities of MPs in mammals, plastic microbeads with a certain size were used as the exposure materials (Zheng et al., 2021; Hou et al., 2021; Jin et al., 2021). However, due to mechanical wear, chemical oxidation, ultraviolet (UV) radiation and biological effects, most MPs in the environment age over time (Salehi et al., 2018). Ageing leads to different physical properties of MPs and might further induce altered health effects compared to pristine MPs. At present, UV irradiation and ozone oxidation were two common ways to simulate the ageing process of MPs in the laboratory studies (Liu et al., 2019b, 2021). Ozone-treated MPs were found to be in line with the characteristics of MPs in the environment

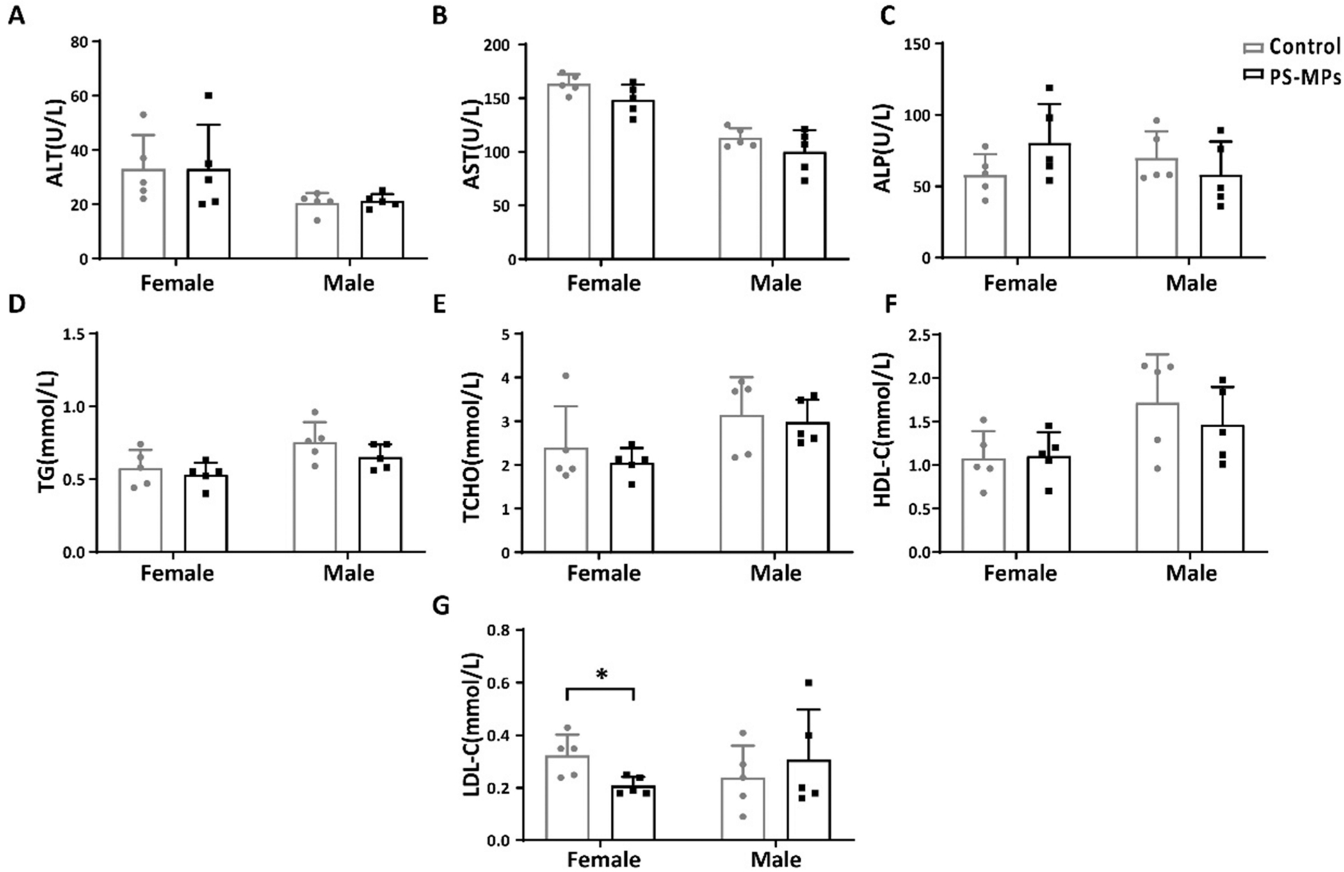

Fig. 3. Blood biochemical results including ALT (A), AST (B), ALP (C), TG (D), TCHO (E), HDL-C (F) and LDL-C (G) in aged PS-MPs treated and control male and female mice. * : $P < 0.05$.

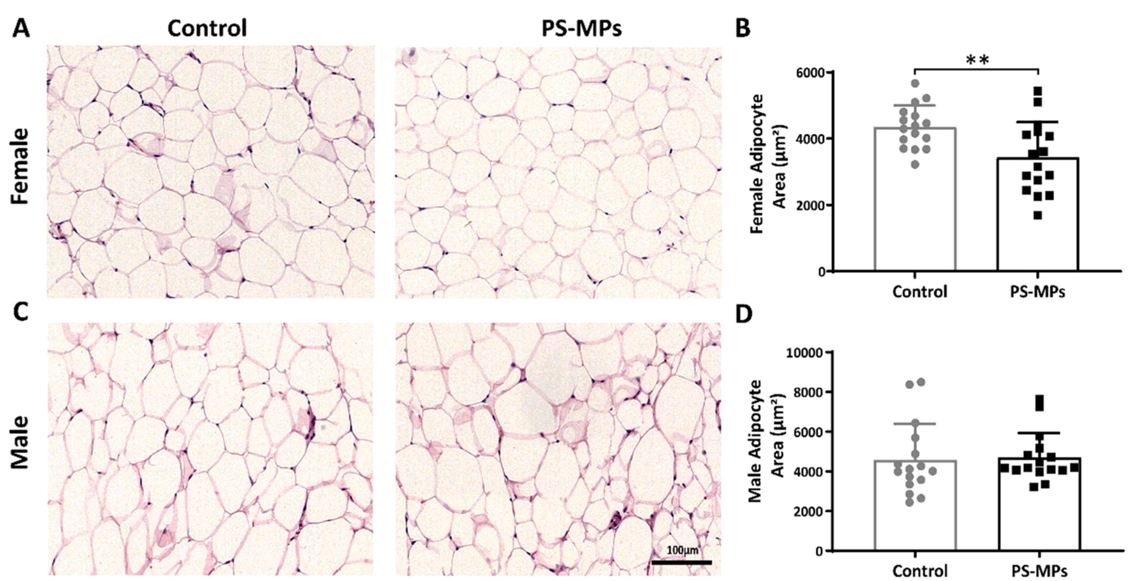

Fig. 4. H&E-stained abdominal fat images (A&C) and adipocyte size comparison in females and males (B&D). * *: $P < 0.01$.

(Guimarães et al., 2021). Unlike plastic microbeads which had spherical shapes, smooth surfaces and uniform sizes (Sun et al., 2021; Jin et al., 2021), both grinded PS-MPs and aged PS-MPs in our study showed irregular shapes, rough surfaces and uneven sizes. Ozone treatment increased oxidation of PS-MPs and decreased their sizes.

The toxicities of PS-MPs may come from their physical properties or from the PS-MPs leachate. In the latter case, PS-MPs act as vectors, carrying various additives, including stabilizers, antioxidants. Studies have already proved ageing could accelerate exudation of additives (Chen et al., 2021a; Wang et al., 2020a). However, PS-MPs oral exposure to mice lasted for 28 days in our study. PS-MPs migrated in the mice body and entered liver, spleen, ovaries et al. Since exudation of additives

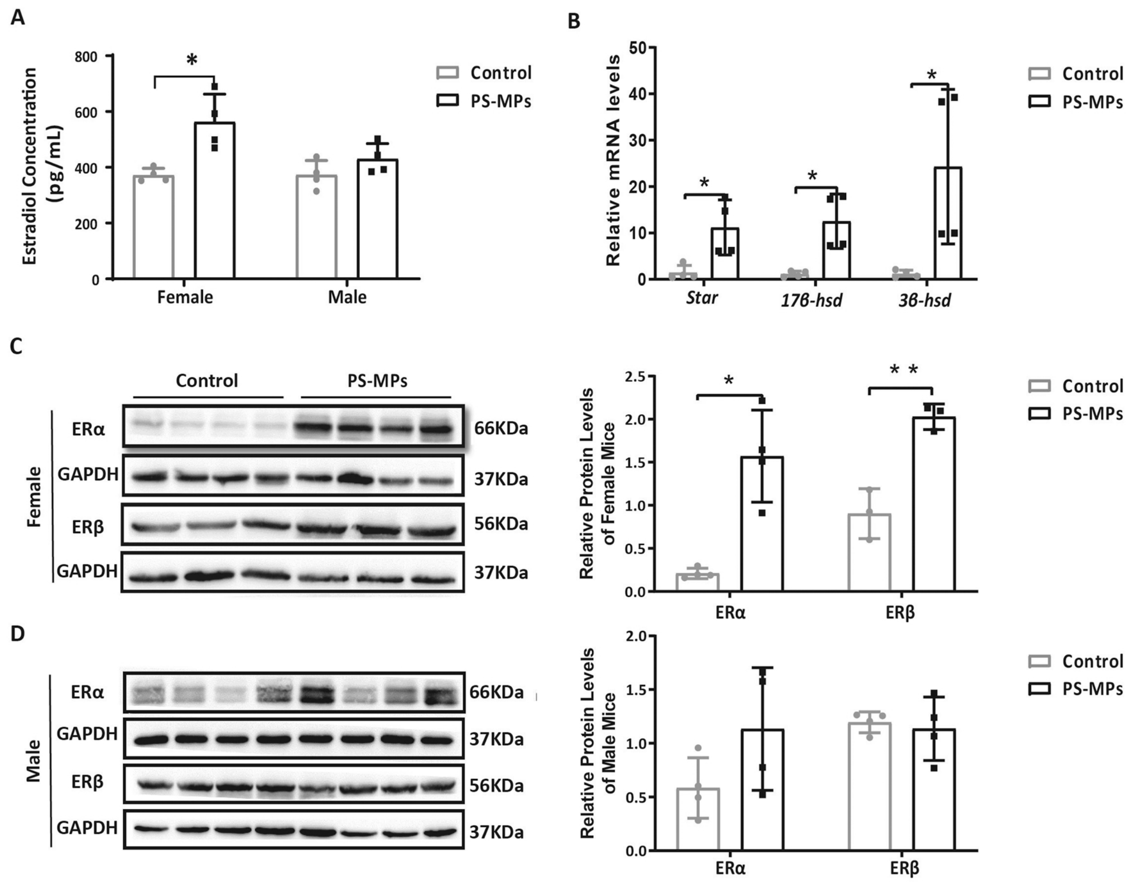

Fig. 5. E2 production in ovary and E2 receptor expressions in liver were detected. (A): Serum E2 concentration in the female and male mice (three replicates). (B): Relative mRNA expression levels of *Star*, *17β-hsd* and *3β-hsd* in the ovary (four replicates for each). (C&D): Relative protein expression levels of ERα, ERβ and GPR30 in the female liver and male liver. * : $P < 0.05$, * *: $P < 0.01$.

from PS-MPs is a continuous process, it is hard to measure how much leachate was leaked into one specific organ and to evaluate the toxicity of leachate to one specific system. Besides, PS-MPs acted as a migrating reservoir of additives. Additives leaking slowly from PS-MPs went through different metabolism process compared to additives alone. Therefore, in this study, we focused more on the combined toxicities of PS-MPs themselves and their leaked additives.

Methods on MPs detection are still developing and confocal laser scanning microscope is commonly used to detect fluorescence-stained MPs inside mammals (Deng et al., 2017; Jin et al., 2021; Han et al., 2021; Im et al., 2021). However, fluorescent MPs are usually made with pristine MPs. There is currently no technical means to embed fluorescent materials in the grinded PS-MPs. Thus, in our study, the LabRAM HR evolution Raman spectrometer with high accuracy was used for MPs detection. After being exposed to aged PS-MPs (<50 μm) for 28 days, PS-MPs were detected in female liver, ovary and spleen and male liver, testis and spleen. Spleen contained comparatively more PS-MPs in males. The possible reason may be that the spleen is the largest lymphatic organ in mammals and contains a large number of macrophages and lymphocytes, which can absorb and phagocytose foreign particles in the blood (Liu et al., 2022). Considering that only 45 mg-piece of tissue were digested for MPs detection, there might be more PS-MPs in the same tissue. However, MPs accumulation is determined not only by tissue volume, but blood flow velocity, vascular distribution, and tissue affinity to the MPs et al. Thus MPs is not evenly distributed in the tissue. The detection of MPs in the tissue can only indicate the presence of MPs, but cannot reflect the accurate quantity of MPs in the tissues and organs.

Oral exposed PS-MPs go in the digestive tract first and lead to inflammation and oxidative stress. The detection of PS-MPs in tissues means that microplastics can cause health effects not only through overall inflammation and oxidative stress, but also through directly entering the specific tissues and leading to local physical damages. Therefore, toxicological effects of MPs and mechanism underlying need to be updated with both biological and physical properties considered.

In our study, the decreased abdominal fat/body coefficient, adipocyte size and LDL-C level after aged PS-MPs exposure in the female but not male mice indicated that aged PS-MPs exposure might inhibit lipid synthesis only in female mice. These sex differences made sex hormones the focus of research. Studies showed that E2 as the main circulating estrogen might bind to the nuclear receptors ERα and ERβ as well as GPR30 to suppress hepatic lipogenic programs (Tian et al., 2019). In the current study, compared to the control, the serum E2 level and the mRNA expression levels of the key enzymes of E2 synthesis including *Star*, *17β-hsd* and *3β-hsd* (Singh and Krishna, 2010; Yao et al., 2020) in ovary were up-regulated only in the female, but not male aged PS-MPs treated group. Why PS-MPs affects the same enzymes only in females but not males are not clear. It may involve local molecule interaction. More investigations need to be done to explore the mechanism on how the MPs stimulate E2 production and release.

The protein expressions of ERα and ERβ also increased in female liver significantly after aged PS-MPs exposure. E2 was found to activate AMPK pathway to suppress the expression of genes related to hepatic lipid synthesis (Pedram et al., 2013). AMPK is a key molecule in the regulation of biological energy metabolism and a major regulator of lipid and glucose metabolism (Ke et al., 2018; Jeon, 2016). Activated AMPK played a key role in regulating lipid metabolism by stimulating a variety of downstream genes (Smith et al., 2016). p-AMPK inhibits

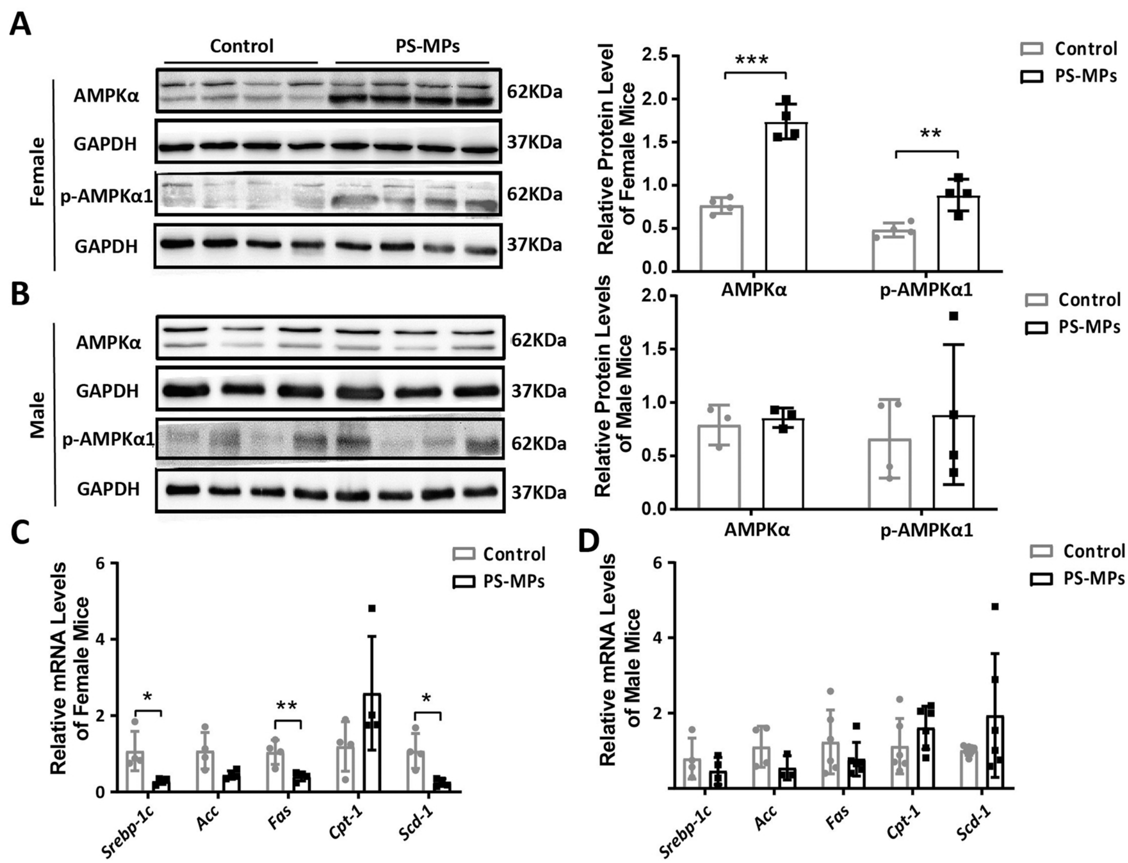

Fig. 6. The expressions of key elements in AMPK pathway were detected. (A&B): Relative protein expression levels of AMPKα and p-AMPKα1 in the female liver and male liver. (C&D): Relative mRNA expression levels of *Srebp-1c*, *Acc*, *Fas*, *Cpt1* and *Scd1* in the female liver and male liver (four replicates for each). * : $P < 0.05$, * *: $P < 0.01$, * ** : $P < 0.001$.

Table 1
Differential metabolites between aged PS-MPs group and control group and the pathway enriched.

	Metabolites	Fold Change	Pathway Enriched	P-value
Female	Biotin	1.791	Biotin metabolism	0.028*
	3'-AMP	0.723	Pyrimidine Metabolism	0.198
			Tryptophan Metabolism	0.202
			Bile Acid Biosynthesis	0.219
	2-Piperidinemethanol	0.403		
	N-Acetylleucine	0.372		
	Purine	1.653		
Male	Xanthosine	1.674	Purine Metabolism	0.478
	L-Leucine	1.207	Valine, Leucine and Isoleucine Degradation	0.407
	3-Methylhistidine	1.989	Histidine Metabolism	0.3
			Beta-Alanine Metabolism	0.232
	6-Hydroxynicotinic acid	1.548		
	3-Pyridylacetic acid	1.548		
	Hyodeoxycholic acid	4.498		
	2-Piperidinemethanol	2.516		
	Norvaline	1.350		
	N-Acetylvaline	1.619		
	Ursodeoxycholic acid	3.244		
	2-Hydroxycaproic acid	0.570		

* : $P < 0.05$.

lipogenesis by inhibiting *Srebp-1c* production (Li et al., 2011) which promotes liver fatty acid biosynthesis by up-regulating adipogenesis related genes, such as *Acc*, *Fas* and *Scd1* (Wang et al., 2019, 2020b). *Cpt1* is involved in the β-oxidation of fatty acids to prevent excessive lipid accumulation in the liver (Lee et al., 2017). In our study, after being exposed to aged PS-MPs, the expression of AMPKα and p-AMPKα1 were up-regulated, and *Srebp-1c*, *Fas* and *Scd1* were down-regulated only in the female liver. Taken together, these results indicated that E2 might bind to nuclear receptors (ERα and ERβ) in liver and further activate AMPK pathway to suppress fatty acid biosynthesis in the female liver, which was consistent with previous research (Pedram et al., 2013).

Liver metabolomic profiling showed that the biotin level increased significantly after being exposed to aged PS-MPs in the female liver. Although it further confirmed the sex-specific effects induced by aged PS-MPs exposure, whether biotin is also regulated by E2 needs solid evidence. Biotin partially restores mitochondrial function, normalizes cholesterol and fatty acid synthesis, and prevents apoptosis and autophagy (Sghaier et al., 2019). It has also been reported that biotin, could activate AMPK signaling pathway (Moreno-Méndez et al., 2019), regulates the expression of genes critical in intermediate metabolism (Fernandez-Mejia, 2005). Therefore, biotin might act as an intermediate link in the regulation of E2 to AMPK pathway (Fig. S4).

Lu Lab and Deng lab reported that PS-MPs could induce hepatic lipid metabolism disorder in male mice (Lu et al., 2018; Deng et al., 2017) which seems to be inconsistent with our results. However, these labs did not compare the effects between genders. In our study, male lipid metabolism was comparatively low only compared to females after PS-MPs exposure. Lipid metabolism disorder in males might be observed when PS-MPs exposure increases. What's more, the difference among studies might result from exposure materials. These labs used pristine

MPs instead of aged MPs.

It is interesting to know whether this sex-specific health effects were from PS-MPs themselves or from their leachate. The sex-specific health effects were not reported in mammals before. We could hypothesize that PS-MPs disrupt steroidogenic pathway but leachate from PS-MPs could exacerbate this effect. More studies need to be done to testify this hypothesis.

5. Conclusion

In conclusion, aged PS-MPs could promote the production and release of E2 which then binds to nuclear receptors (ERα and ERβ) to activate the AMPK signaling pathway, inhibiting liver lipid synthesis in female, but not male mice. The sex differences in aged PS-MPs-exposure-induced effects on liver lipid metabolism provide new insights into the potential sex-specific health effects of environmental MPs pollution.

CRediT authorship contribution statement

Xiaona Yang: Methodology, Investigation, Data curation, Writing - original draft, Writing - review & editing. **Jin Jiang:** Methodology, Investigation, Data curation, Validation. **Qing Wang:** Methodology, Investigation, Formal analysis. **Jiawei Duan:** Methodology, Formal analysis. **Na Chen:** Validation. **Di Wu:** Project administration, Supervision, Writing - review & editing. **Yankai Xia:** Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2022.114105.

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